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Marketing Mix Modeling Platform

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DS223 — Marketing Analytics

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1. Problem Statement

Marketing teams at companies of all sizes face a persistent and expensive challenge: they cannot reliably determine which advertising channels are actually driving revenue. Budgets are routinely allocated across television, paid social media, organic search, display advertising, print, and email — yet decision-makers lack a scientific, repeatable methodology to attribute revenue to specific channels, quantify diminishing returns, or determine the optimal mix of spend going forward.

1.1 Problem Area

This project sits at the intersection of Marketing Channel Attribution and Budget Optimisation. The core challenge belongs to the broader domain of marketing effectiveness measurement — a field where decisions worth millions of dollars are routinely made on intuition, last-click attribution heuristics, or vendor-supplied metrics that carry obvious conflicts of interest.

1.2 Preliminary Research Findings

Industry Scale & Wasted Spend: Global digital ad spend reached approximately \$602 billion in 2023. Commerce Signals, drawing on roughly 60 independent studies, estimates that 40% of all media spend is wasted — primarily because marketers cannot accurately measure the sales impact of their advertising (Navarro, 2025).

Attribution Gap & Regulatory Pressure: Last-click attribution systematically undervalues top-of-funnel channels while overvaluing bottom-funnel tactics that simply capture existing demand. Privacy regulations including GDPR and CCPA are accelerating the shift away from user-level tracking toward aggregate methodologies such as MMM (Sushil, 2026).

1.3 Specific Problem Definition

Problem Statement: *Marketing managers at small-to-mid-size organisations lack an accessible, automated platform that applies Marketing Mix Modeling to their historical weekly spend and revenue data in order to produce statistically grounded channel ROI estimates, reveal saturation and carryover effects, and recommend an optimal budget reallocation — resulting in significant wasted ad spend and missed revenue opportunities.*

The problem is specific (channel attribution and budget optimisation), measurable (ROI per channel, revenue lift from reallocation), relevant (affects every company that spends on paid media), and tractable via established statistical methodology.

2. Proposed Solution

We propose to build a web-based Marketing Mix Modeling Platform that combines a proven statistical methodology with a five-microservice software architecture. The platform ingests weekly marketing spend and revenue data, applies adstock and saturation transformations, trains a multivariate regression model, and surfaces channel ROI estimates alongside an interactive budget optimiser through a clean, role-appropriate dashboard.

2.1 Data Collection

The platform will accept user-uploaded CSV files containing weekly time-series data. Required fields include: `week_start_date`, one or more `spend_[channel]` columns (e.g., `spend_tv`, `spend_social`, `spend_search`), and a revenue column. Optional covariates include seasonality indicators, promotional flags, and economic index data. A guided upload wizard will validate schema, detect outliers, and surface data-quality warnings before model training begins.

2.2 Analytical Methodology

Adstock Transformation

Models the carryover (lagged) effect of advertising spend. Each channel's spend is convolved with a geometric decay function, allowing the model to capture the fact that a TV ad aired in week 1 continues to influence purchasing in weeks 2–4.

Saturation / Diminishing Returns

A Hill or logistic saturation curve is applied to each transformed spend variable to capture the non-linear relationship between spend and response — the 'diminishing returns' effect observed when a channel is over-invested.

Ridge / Bayesian Linear Regression

After transformation, a regularised (Ridge) or Bayesian linear regression is trained on the full time series. Coefficients directly encode channel contribution to revenue. Bayesian formulation additionally yields posterior uncertainty intervals for each channel ROI, supporting better decision-making under uncertainty.

Budget Optimisation

Given a fixed total budget and the trained model, a constrained optimisation (scipy minimise with bounds) finds the spend allocation across channels that maximises predicted revenue while respecting per-channel minimum and maximum constraints.

2.3 System Architecture

The platform is decomposed into five loosely-coupled microservices, each with a dedicated codebase, database schema, and deployment unit:

Service	Technology	Responsibility
Data Ingestion	Python / FastAPI	CSV upload, validation, raw storage
Data Pipeline	Apache Airflow / dbt	Cleaning, feature engineering, adstock
Modelling Engine	Python / scikit-learn	MMM training, ROI scoring, optimiser
REST API Gateway	FastAPI / PostgreSQL	Unified API surface, auth, caching
Web Dashboard	React / Recharts	Charts, budget slider, export

All five microservices are containerised using Docker. Each service has its own Dockerfile, and the entire platform is orchestrated locally using docker-compose. This means any team member can spin up the full application with a single command (docker-compose up) without manually installing dependencies. It also ensures the platform behaves the same way on every machine.

2.4 Implementation Plan/ Roadmap

- Sprint 1 (Week 1): Repo setup, Dockerfile creation for each service, docker-compose configuration, CI/CD skeleton, database schema, CSV upload endpoint..
- Sprint 2 (Week 2): MMM model training, cross-validation, ROI output.
- Sprint 3 (Week 3): Budget optimizer, REST API v1, Postman collection, frontend scaffold.
- Sprint 4 (Week 4): React dashboard, API integration, testing, demo dataset, final presentation.

3. Expected Outcomes

Transparency

Marketing managers gain a clear, quantitative breakdown of how much each channel contributes to weekly revenue, replacing opaque vendor dashboards with a model they own and can audit.

Budget Efficiency

By identifying over-invested channels experiencing saturation and under-invested channels with remaining headroom, the optimiser is projected to improve blended marketing ROI by 15–30% on benchmark datasets used for evaluation.

Accessible MMM

The platform democratises a methodology previously available only to large brands with six-figure analytics budgets, making it accessible to SMEs through a self-serve web interface.

Privacy Compliance

Because MMM operates on aggregate weekly data rather than individual-level user tracking, the platform is inherently compliant with GDPR and CCPA.

Definition of done

The MVP is complete when all five microservices start via docker-compose up, the pipeline populates the database from the datasets, the model produces channel coefficients, the dashboard renders the ROI chart and optimizer, and a live demo can be presented in under 10 minutes.

4. Evaluation Metrics (KPIs)

KPI	Definition	Target
Model R ²	Variance in revenue explained by the trained MMM	≥ 0.80
MAPE	Mean Absolute Percentage Error on hold-out test weeks	$\leq 12\%$
ROI Stability	Std. dev. of channel ROI across 5-fold CV folds	$< 15\%$
Optimiser Lift	% revenue increase predicted by optimised vs. actual spend	$\geq 10\%$
Upload-to-Result Time	End-to-end latency from CSV upload to dashboard load	≤ 60 s
API Uptime	System availability measured over demo period	$\geq 99\%$
Container PortabilityFull	platform runs via docker-compose up with zero manual setup	Yes

5. Role Responsibility

- **Product / Project Manager:** Roadmap, coordination, stakeholder alignment
- **Database Developer:** Schema design, storage, query optimisation
- **API / Backend Developer:** REST microservices, orchestration layer
- **Frontend Developer:** Dashboard, budget optimiser, data viz
- **Data Scientist:** MMM model, adstock/saturation, validation
- **Data Pipeline Engineer:** Ingestion, ETL, scheduling, monitoring

6. Prioritized Feature Backlog (MoSCoW)

All project features are prioritized using the MoSCoW framework. Priority determines what must be completed for a passing MVP, what adds meaningful quality, and what is deferred if time runs short. Effort is rated Low / Medium / High based on estimated implementation complexity.

Must have — core MVP, non-negotiable	Should have — strongly recommended	Could have — nice to have if time allows	Post-MVP — out of scope	
Feature	Priority	Effort	Owner	Sprint
PostgreSQL schema — all tables	Must have	Low	DB	S1
Docker Compose — all 5 services	Must have	Medium	DB	S1
CSV ingestion	Must have	Low	Data Engineer	S1
Adstock transformation	Must have	Medium	Data Engineer	S2
Saturation (Hill function) transform	Must have	Medium	Data Engineer	S2
Holiday feature engineering	Must have	Low	Data Engineer	S2
OLS regression model	Must have	Medium	ML Engineer	S2
Channel coefficient + ROI output	Must have	Low	ML Engineer	S2
FastAPI — GET /results endpoint	Must have	Low	Backend Dev	S3
FastAPI — POST /optimize endpoint	Must have	Medium	Backend Dev	S3
React dashboard — ROI bar chart	Must have	Medium	Frontend Dev	S3
React dashboard — response curves	Must have	Medium	Frontend Dev	S3
Budget allocation slider (optimizer UI)	Must have	High	Frontend Dev	S4
Scenario save + comparison view	Should have	High	Frontend Dev	S4
Bayesian model (PyMC) upgrade	Should have	High	ML Engineer	S4
Model run history log in dashboard	Should have	Medium	Backend Dev	S4
Export results to CSV	Could have	Low	Frontend Dev	S4
Multi-market / multi-brand support	Could have	High	All	Post-MVP

7. Conclusion

The proposed MMM Analytics Platform addresses a well-documented, costly gap in the marketing technology landscape. By combining statistically rigorous channel attribution with an accessible web interface and an automated budget optimiser, the project will deliver measurable value to marketing teams while demonstrating the full data-engineering and analytics stack covered in this course. The six-member team brings complementary expertise across product management, database engineering, backend API development, frontend design, data science, and pipeline infrastructure — ensuring every layer of the architecture is owned by a dedicated specialist.

References

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